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Q1. (Preparing Features)

Load your cleaned Ford Car Dataset.

Separate the dataset into Independent Features (X) and Dependent Feature (Y) (price).

Drop the target column from X.

Print the shape of X and Y.

```
import pandas as pd
```

```
df = pd.read_csv('ford.csv')
```

```
X = df.drop('price', axis=1) # Remove target column from features
```

```
Y = df['price']
```

```
print("Shape of X (Independent Features):", X.shape)
```

```
print("Shape of Y (Dependent Feature):", Y.shape)
```

Q2.(One-Hot Encoding)

Identify all categorical columns in X.

Perform One-Hot Encoding using `pd.get_dummies()`.

Convert the encoded data to integer type (`astype(int)`).

Show the first 5 rows after encoding.

```
categorical_columns = X.select_dtypes(include=['object']).columns
```

```
print("Categorical Columns:")
```

```
print(categorical_columns)

X_encoded = pd.get_dummies(X, columns=categorical_columns)

X_encoded = X_encoded.astype(int)

print("\nFirst 5 rows after One-Hot Encoding:")

print(X_encoded.head())
```

Q3.(Feature Scaling)

Apply Standard Scaling on the numerical columns (year, mileage, tax, mpg, engineSize).

Use StandardScaler from sklearn.preprocessing.

Show the first 5 rows of the scaled features.

```
from sklearn.preprocessing import StandardScaler

numerical_columns = ['year', 'mileage', 'tax', 'mpg', 'engineSize']

scaler = StandardScaler()

X_encoded[numerical_columns] = scaler.fit_transform(
    X_encoded[numerical_columns]
)

print("First 5 rows after Standard Scaling:")

print(X_encoded.head())
```

Q4.(Train-Test Split)

Split the preprocessed data into training and testing sets using `train_test_split`.

Use `test_size=0.33` and `random_state=42`.

Print the shape of `X_train`, `X_test`, `y_train`, and `y_test`.

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X_encoded,
```

```
    Y,
```

```
    test_size=0.33,
```

```
    random_state=42
```

```
)
```

```
print("X_train shape:", X_train.shape)
```

```
print("X_test shape:", X_test.shape)
```

```
print("y_train shape:", y_train.shape)
```

```
print("y_test shape:", y_test.shape)
```

Q5.(Building Linear Regression Model)

Train a Linear Regression model using `sklearn`.

Fit the model on the training data (`X_train`, `y_train`).

Print the intercept and coefficients of the model.

```
from sklearn.linear_model import LinearRegression
```

```
lr_model = LinearRegression()
```

```
lr_model.fit(X_train, y_train)
```

```
print("Intercept:", lr_model.intercept_)
```

```
print("\nCoefficients:")  
  
for feature, coef in zip(X_train.columns, lr_model.coef_):  
    print(f"{feature}: {coef}")
```

Q6.(Making Predictions)

Use the trained model to make predictions on the test data.

Store predictions in y_pred.

Print the first 10 predicted values and compare them with the first 10 actual values from y_test.

```
y_pred = lr_model.predict(X_test)  
print("First 10 Predicted Values:")  
print(y_pred[:10])  
print("\nFirst 10 Actual Values:")  
print(y_test.iloc[:10].values)  
print("\nComparison of Predicted and Actual Values:")  
comparison = pd.DataFrame({  
    'Actual Price': y_test.iloc[:10].values,  
    'Predicted Price': y_pred[:10]  
})  
print(comparison)
```

Q7.(Model Evaluation using R² Score)

Calculate the R² score using r2_score(y_test, y_pred).

Interpret the R² value in your own words.

Comment on how well the model is performing.

```
from sklearn.metrics import r2_score
```

```
r2 = r2_score(y_test, y_pred)
```

```
print("R2 Score:", r2)
```

Q8.(Saving the Model)# Import joblib

```
import joblib
```

```
joblib.dump(lr_model, "LR_ford_car.pkl")
```

```
joblib.dump(scaler, "scaler.pkl")
```

```
joblib.dump(X_encoded.columns.tolist(), "columns.pkl")
```

```
print("Model and preprocessing objects saved successfully!")
```

Save your trained model and preprocessing objects using joblib:

The Linear Regression model → "LR_ford_car.pkl"

The StandardScaler → "scaler.pkl"

The list of columns → "columns.pkl"

Optional Task:

Create a ML model using Dataset you used in Assignment 19

```
import pandas as pd
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.linear_model import LinearRegression
```

```
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error
```

```
df = pd.read_csv("Real estate.csv")
```

```
X = df.drop("Y house price of unit area", axis=1)
```

```
y = df["Y house price of unit area"]
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
print("R2 Score:", r2)
print("Mean Absolute Error:", mae)
print("Mean Squared Error:", mse)
print("\nIntercept:", model.intercept_)
print("\nCoefficients:")
for feature, coef in zip(X.columns, model.coef_):
    print(feature, ":", coef)
```